

Eastern Mediterranean University – Computer Engineering Department

CMPE/CMSE/AING231 – Data Structures

Assignment #4

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- Mr. Chimaobi works for the Ministry of Tourism. His duty is to go to Namık Kemal Square, ask each passing tourist which country he/she comes from, and make a list of the countries where tourists come from. Of course, this process can take days or even weeks. That's why Mr. Chimaobi needs a program. This program will keep track of each country that Mr. Chimaobi enters, and then, list each country (once) in alphabetical order starting from the country given by Mr. Chimaobi.

Here are the points you need to pay attention to::

- Your solution should be with a **linked list** data structure.
- As long as Mr. Chimaobi enters, place the countries into a link list. Let's call it the **main linked list**. While inserting into the main list, do not apply any criteria and insert the new country **directly** to the end of the list.
- **When Mr. Chimaobi's entries are finished**, the **system** will do the following:
 - For those countries which are duplicate in your main linked list, it will keep only the first node and delete the others. You must perform this operation in a separate function.
 - Afterward, it will read individual countries, starting with the first node of your main list, and copy them to a second list. The second list should be a **circular linked list** that stores countries in alphabetical order. Therefore, while copying operations, act accordingly.
 - Finally, only the countries, after the one entered by the user, in the **circular** linked list will be displayed in alphabetical order on the screen.

NOTE: If any of the following situations occur, you will get ZERO:

1. You have not uploaded your solution by the deadline.
2. You have produced a solution other than the C language.
3. You have uploaded an incorrect file or a file that cannot be opened.
4. Your solution contains **even a single line of code** that is NOT yours.
5. **You have shared your solution or a part of your solution with someone else!!**